

Classes of Methods

$$u'(t) = f(u)$$

Linear Multistep Methods:

$$\sum_{j=0}^r \alpha_j u_{n+j} = h \sum_{j=0}^r \beta_j f(u_{n+j})$$

Formula for u_{n+r}
Implicit if $\beta_r \neq 0$

Runge-Kutta Methods: (multi-stage, one step)

$$y_i = u_n + h \sum_{j=1}^s a_{ij} f(y_j)$$

$$u_{n+1} = u_n + h \sum_{j=1}^s b_j f(y_j)$$

Explicit if $a_{ij} = 0$
for $j \geq i$.

"Diagonally implicit" if $a_{ij} = 0$
for $j > i$.

$$1 \leq i \leq s$$

Vectorized notation:

$$Y = u_n e + h A F$$

$$u_{n+1} = u_n + h b^T F \quad \text{Scalar}$$

$$e = [1, 1, \dots, 1]^T \in \mathbb{R}^s$$

$$F = [f(y_1), f(y_2), \dots, f(y_s)]^T$$

General Linear Methods (multistep, multi-stage)
(multivalued)

$$y_i = \sum_{j=1}^r w_{ij} y_j^{[n-1]} + h \sum_{j=1}^s a_{ij} f(y_j) \quad 1 \leq i \leq s$$

$$y_i^{[n]} = \sum_{j=1}^r v_{ij} y_j^{[n-1]} + h \sum_{j=1}^s b_{ij} f(y_j) \quad 1 \leq i \leq r$$

Additive/Split Methods

Solve $u'(t) = f(u) + g(u)$

If f and g have very different properties, we may want to apply different techniques to each.

- Operator splitting/fractional step
- Implicit-explicit
- Exponential

Multiderivative Methods

(can be combined with multistage or multistep)

For example

$$y_i = u_n + h \sum_j a_{ij} f(y_j) + h^2 \sum_j \hat{a}_{ij} \dot{f}(y_j) \quad \dot{f}(y) = \frac{d}{dt} f(y)$$

$$u_{n+1} = u_n + h \sum_j b_j f(y_j) + h^2 \sum_j \hat{b}_j \dot{f}(y_j)$$

$$\frac{d}{dt} f(y(t))_j = \sum_i \frac{\partial f_j}{\partial y_i} \frac{dy_i}{dt}$$

$\underbrace{\frac{dy_i}{dt}}_{f(y)_i}$

Becoming more popular due to auto. diff.

Stability

Zero-stability: as $h \rightarrow 0$ (required for ODE convergence)

Absolute stability: Not required for ODE convergence, but is for PDE convergence.

Recall: $U_t = F(U, U_x, U_{xx}, \dots)$

disc. in space $\Rightarrow \underbrace{u'(t) = f(u)}_{\Delta x}$

As we refine the space grid, the Lipschitz const. of f must grow.

By "PDE convergence" we mean convergence as both h and $\Delta x \rightarrow 0$.

Region of Absolute Stability

Comes from considering $u'(t) = \lambda u(t)$
 $f(u) = \lambda u$

RKMs:

$$Y = u_n e + h A F \quad F = \lambda Y \quad z = h \lambda$$
$$u_{n+1} = u_n + h b^T F$$

$$Y = u_n e + z A Y \Rightarrow Y = (I - z A)^{-1} e u_n$$

$$u_{n+1} = u_n + h \lambda b^T Y = (1 + z b^T (I - z A)^{-1} e) u_n$$

$$u_{n+1} = R(z)u_n \quad R(z): \text{Stability function}$$

$$\text{Example: 4th order RK} \quad R(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}$$

$$S = \{z \in \mathbb{C} : |R(z)| \leq 1\}$$

LMMs:

$$\sum_{j=0}^r \alpha_j u_{n+j} = h \sum_{j=0}^r \beta_j f(u_{n+j}) \quad f(u) \rightarrow \lambda u$$

$$\sum_{j=0}^r \alpha_j u_{n+j} = z \sum_{j=0}^r \beta_j u_{n+j} \Rightarrow \sum_{j=0}^r (\alpha_j - z\beta_j) u_{n+j} = 0$$

$$\text{Solution: } u_n \rightarrow \zeta^n$$

$$\sum_{j=0}^r (\alpha_j - z\beta_j) \zeta^j = 0$$

$$\underbrace{\sum_{j=0}^r (\alpha_j - z\beta_j) \zeta^j}_{\pi(\zeta; z)} = 0 \quad \text{Stability function}$$

$$S = \{z \in \mathbb{C} : \text{The roots of } \pi \text{ satisfy the root condition}\}$$

The root condition: all roots satisfy $|\zeta| \leq 1$ and
if $|\zeta| = 1$, ζ is a simple root.

On the boundary of S , we have $\zeta = e^{i\theta}$ for
some root. So

$$z = \frac{\sum \alpha_j e^{i\theta j}}{\sum \beta_j e^{i\theta j}}$$

boundary locus curve
 $0 \leq \theta \leq 2\pi$

$$\prod (\mathcal{P}_j; z) = (\mathcal{P} - \mathcal{P}_1)(\mathcal{P} - \mathcal{P}_2) \cdots (\mathcal{P} - \mathcal{P}_r)$$

$$\mathcal{P}_i \neq \mathcal{P}_j \text{ for } i \neq j \quad (\text{simple root})$$

GLMs

$$F \Rightarrow \lambda Y$$

$$Y = WY^{[n-1]} + hAF$$

$$Y = WY^{[n-1]} + zAY$$

$$Y^{[n]} = VY^{[n-1]} + hBF$$

$$Y = (I - zA)^{-1} WY^{[n-1]}$$

$$Y^{[n]} = \underbrace{\left[V + zB(I - zA)^{-1}W \right]}_{\text{Eigenvalues must satisfy the root condition}} Y^{[n-1]}$$

Eigenvalues must satisfy
the root condition